2019 Term B solutions

Question 1. (6 points)

Explicitly write two non-empty sets $A, B \subseteq \{\emptyset, \{\emptyset\}\}$ such that :

$A \subseteq P(A)$ and $B \not\subseteq P(B)$

$$A = \{\emptyset\}, \qquad B = \{\{\emptyset\}\}$$

Question 2. (6 points)

Which subset of the following four predicates are true in $\mathbb{R}$?

1. $\forall x(\exists a, b, c(ax^2 + bx + c = 0))$
2. $\exists x(\forall a, b, c(ax^2 + bx + c = 0))$
3. $\forall a, b, c(\exists x(ax^2 + bx + c = 0))$
4. $\exists a, b, c(\forall x(ax^2 + bx + c = 0))$

1,4

Question 3. (6 points)

Explicitly write a function $f: P(\mathbb{N} \setminus \{7\}) \to P(\mathbb{N}) \setminus P(\{7\})$ which is not 1 - 1.

$$f(A) = \begin{cases} A & A \neq \emptyset \\ \{1,7\} & A = \emptyset \end{cases}$$

Question 4. (6 points)

You want to choose a committee of three people with the roles of a chairman, a vice president and a secretary (a person can only have one role). Given a group of 4 men and 6 women, how many such committees have both genders?

$$10 \cdot 9 \cdot 8 - 4 \cdot 3 \cdot 2 - 6 \cdot 5 \cdot 4 = 576$$

Question 5. (6 points)

How many **integer** solutions are there to the equation:

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 15$$

Where $\forall i \;\; 0 \leq x_i \leq 5$?

$$e_0 = \binom{15+6-1}{5} - \binom{6}{1}\binom{9+6-1}{5} + \binom{6}{2}\binom{3+6-1}{5}$$

Question 6. (6 points)

In how many ways can you organize 6 different books on 5 shelves, such that the top, bottom and middle shelves each have exactly one book (order within a shelf does not matter).

$$6 \cdot 5 \cdot 4 \cdot 2^3$$

Question 7. (6 points)

Let a, b, c be 3 natural numbers such that $a + b + c = 19$.

Prove that one of the 3 numbers is greater or equal to 7.

Guidance: Briefly explain (4 lines max) using the pigeonhole principle.

State which are the pigeons, which are the pigeonholes and why the claim is true given what you have defined.

We define the pigeons to be 19 units (i.e., the sum).

The pigeonholes are the three variables $a, b, and\ c$

By the extended pigeonhole principle *there is a pigeonhole with at least* .$\left\lceil \frac{19}{3} \right\rceil = 7$

Pigeons, and therefore there exists a number that is greater or equal to 7

Question 8. (6 points)

Write the following expression in a concise way (using 5 characters max).

* Notice the sigma limits.

$$\sum_{k=0}^{9} \binom{19}{k}$$

$$2^{18}$$

Question 9. (6 points)

Let R be a relation on $\mathbb{N}$ defined as follows:

aRb iff $\frac{b^2}{2a} \in \mathbb{N}$

The relation is **not** : Reflexive, Anti Reflexive, Symmetric nor Anti Symmetric.

Explicitly write pairs which serve as a counter example for each of the properties.

Not-reflexive, counterexample: $(3,3) \notin R$.

Not anti-reflexive, counterexample: $(4,4) \in R$.

Not symmetric, for example, $(1,2) \in R$ but $(2,1) \notin R$.

Not anti-symmetric, for example: $(4,8) \in R$ and $(8,4) \in R$ but $4 \neq 8$.

Question 10. (6 points)

Let $A = \{1,2,3,4,5,6\}$ be a set.

And let $\mathrm{S} = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$ and $\mathrm{T} = \{(2,3), (3,2), (1,6)\}$ be two sets.

How many partial order relations R on A are there such that $R \subseteq S \cup T$?

$$12$$

The relation can be either weak (includes S) or strong (does not include S) – 2 options.

As the order is anti-symmetric, for the pairs (2,3), (3,2) we have 3 options: taking one of them or none. The pair (1,6) can be in or not – two options. In total: $2 \cdot 3 \cdot 2 = 12$

Question 11. (6 points)

Let $A = \{2^n \cdot 3^m \mid (n, m \in \mathbb{N}) \wedge (n,\ m > 0) \wedge (n, m \leq 3)\}$

We define a relation R on A as follows:

xRy iff the highest power of 6 which divides x , is the same as the highest power of 6 which divides y.

Write the equivalence classes of R.

$$\mathrm{A/R} = \{ [6]_{\mathrm{R}}, [36]_{\mathrm{R}}, [216]_{\mathrm{R}}\}$$

Question 12. (6 points)

Ronda, Donna, Tiff, Grace, Norah and Sylvia are looking for a partner to join them in the Amazing Race game show. They can choose from: Gabe, Don, John, Aaron, Mike and Ethan.

Ronda **won't** go with: Ethan and Gabe.
Donna **won't** go with: Don, Gabe and Ethan.
Tiff **won't** go with: Mike, Gabe and Ethan.
Grace **won't** go with: Don, Aaron and Mike.
Norah **won't** go with: Mike, Gabe, Aaron and Ethan.
Sylvia **won't** go with: Gabe, Aaron and Ethan.

Is there a matching where all the girls are happy (won't be matched with someone they don't want)?
If so, write the matching, if not, briefly explain why.

There is not such matching. The preference set of 5 women includes 4 men, in contradiction to the necessary conditions in Halls theorem.

Question 13. (14 points)

Let w be a binary word of length n. Let $w(i)$ indicate the i'th bit in w.

We define w^C the complement word of w : $w^C(i) = 1 - w(i)$

We define w^R as the reverse word of w : $w^R(i) = w(n + 1 - i)$

For example, if w is 10010, then w^C is 01101 and W^R is 01001.

Prove by induction that for every $n \geq 1$, every binary word w of length n satisfies $(w^C)^R = (w^R)^C$

<u>Solution:</u>

By induction on n.

Base case: for $n = 1$, w is either 0 or 1. We will show correctness for each case:

$$w = \text{"1"}: \quad (w^R)^C = 1^C = 0, \qquad (w^C)^R = 0^R = 0$$
$$w = \text{"0"}: \quad (w^R)^C = 0^C = 1, \qquad (w^C)^R = 1^R = 1$$

Induction Step: Assume the claim holds for words of length $n \geq 1$, that is, that for any word u of length n: $(u^C)^R = (u^R)^C$, and we will show correctness for words of length $n + 1$. Let w be a binary word of length $n + 1$, and we next show $(w^C)^R = (w^R)^C$.

Denote the bits of w by: $w_1 w_2 \dots w_n w_{n+1}$, and the prefix $w_1 w_2 \dots w_n$ denote by u.

It therefore holds that:

$$(w^R)^C = ((u\, w_{n+1})^R)^C = (w_{n+1}\, u^R)^C = (w_{n+1})^C\, (u^R)^C$$
$$(w^C)^R = ((u\, w_{n+1})^C)^R = (u^C\, w_{n+1}{}^C)^R = (w_{n+1})^C\, (u^C)^R$$

By the induction assumption, $(u^C)^R = (u^R)^C$, and therefore we can conclude that

$$(w^R)^C = (w_{n+1})^C\, (u^R)^C = (w_{n+1})^C\, (u^C)^R = (w^C)^R$$

Question 14. (14 points)

Let $f: A \to A$ be a total function, and $g: A \to A$ a total bijective function.

Prove that if $f \circ f \circ g$ is 1 - 1 then f is 1 - 1 as well.

<u>Solution:</u>

Assume toward contructions that $f \circ f \circ g$ is 1-1 but f is not 1-1.

Since f is not 1-1, there exists $x_1 \neq x_2$ such that $f(x_1) = f(x_2)$.

Since g is a bijective, it is onto, and therfore there exists $y_1, y_2 \in A$ such that $g(y_1) = x_1$ and $g(y_2) = x_2$. Moreovere, since $x_1 \neq x_2$ and$g$ is 1-1 (bijective), then $y_1 \neq y_2$.

Hence, when we apply $f \circ f \circ g$ on y_1 and y_2 we get:

$$f \circ f \circ g(y_1) = f\big(f(x_1)\big),$$
$$f \circ f \circ g(y_2) = f\big(f(x_2)\big),$$

Since $f(x_1) = f(x_2)$ we conclude that

$$f \circ f \circ g(y_1) = f\big(f(x_1)\big) = f\big(f(x_2)\big) = f \circ f \circ g(y_2)$$

In contrudiction to the assumption that $f \circ f \circ g$ is 1-1.